

Vol 15 Chapter 10 — Calibration Stack

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 10 — Calibration Stack

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

BST's calibration stack — 23+ Cal standing calibrations + 2 candidate (as of 2026-05-24) + 25 STANDING + 1 CANDIDATE + 1 META methodology layers (per Methodology Index v0.5) — accumulates one layer per detected failure mode, building rigor incrementally from L1 (Quaker discipline) through L21+ (Cal #22 v0.2 absorption-pipeline discipline), with each calibration triggered by a specific incident and preserved as a permanent rule.

Level 2 — graduate-physicist precision

10.1 Calibration trigger pattern

A calibration arises when: 1. An incident occurs (wrong figure, premature ratification, drift, etc.) 2. Cal (or self-audit) flags the failure mode 3. Team discusses root cause 4. Calibration filed as standing rule 5. Methodology stack layer added

10.2 Full calibration list (as of 2026-05-24)

#	Layer	Trigger
1-9	L1-L9 foundational	various early incidents
10	STRUCTURALLY VERIFIED tier	Cal #66

#	Layer	Trigger
11	RIGOROUSLY CLOSED tier	Cal #77
12-15	Phase 2 (M2C2, F1-F4, B1-B4)	various
16	PCAP formalized	Cal #65
17	External register	Cal #19
18	PCAP-rate transcription	Cal #22
19	Substance-floor	Cal #23 (Saturday 2026-05-23)
20	Cross-volume audit	Cal #24 (Saturday 2026-05-23)
21	Cal #22 v0.2 absorption-pipeline (extension)	Cal #119 self-catch (Sunday 2026-05-24)
22	BST-Primary-Target Forward-Derivation Discipline	Cal #121 4th instance trigger (Sunday 2026-05-24 PM) — promoted CANDIDATE→STANDING per Casey directive
23 (per Cal MIndex v0.6)	Additional Cal-tracked layer	Cal Methodology Index v0.6 documents
24 (per Cal MIndex v0.6)	Additional Cal-tracked layer	Cal Methodology Index v0.6 documents
25	Falsifier-Threshold Discipline ($3\sigma/5\sigma$ statistical thresholds for empirical tests)	Sunday 2026-05-24 PM — promoted CANDIDATE→STANDING per Cal #124 cross-promotion note + Keeper acceptance per audit-chain delegation; Elie Toy 3520 photon-pair thresholds (75K/169K/469K for $2\sigma/3\sigma/5\sigma$) provided operational anchor
CANDIDATE	Readiness-label	Cal #26 (Saturday candidate)

Authoritative count per Cal Methodology Index v0.6: 25 STANDING + 1 CANDIDATE (#26) + 1 META (#99) = 27 numbered methodology layers (after #25 + #27 promoted Sunday afternoon).

(See notes/referee_objections_log.md for full incident log including Cal #119

Vol 12-15 cold-read.)

10.3 Recent calibrations (Friday May 22 + Saturday May 23 + Sunday May 24)

Friday May 22: - #18 premature methodology-tier conflation (Lyra T2419 within-session self-correction) - #19 forecast-arithmetic vs ratified-state (external register) - #20 timestamp drift (CI temporal projection ~10-15 min forward) - #21 empirical vs mechanism (Grace K141 honest FAIL preserved) - #22 PCAP-rate transcription error class ($T_{190} 24/\pi^2$ precision misattribution)

Saturday May 23: - #23 substance-floor (every chapter needs minimum substantive content) - #24 cross-volume audit (8-dimension consistency between volumes) - #25 CANDIDATE falsifier-threshold (when do thresholds get formalized?) - #26 CANDIDATE readiness-label (when is content marked v1.0-ready?)

Sunday May 24: - Cal #119 self-catch Cal #22 v0.2 absorption-pipeline applied to Cal: my own Cal #108 inherited $(1/3)^{19} \approx 8.6 \times 10^{-10}$ from Lyra T2467+T2468 v0.2 §6 without recomputing; correct $(1/3)^{18} \approx 2.6 \times 10^{-9}$ with $11+7=18$ max-promotable. Vol 14 Ch 8 + Vol 15 Ch 8 Theorem inherited the same error. All corrected in v0.4 absorption pass. - Cal #121 cold-read on Lyra DCCP v0.3: demoted “RIGOROUS theorems” claim to STRUCTURALLY VERIFIED CANDIDATE per Cal #66 + Cal #77. Two HIGH-priority flags: Lemma DCCP-1.5 cardinality assertion (Wallach infinite K-types + K59 7-step + T2447 137 CAP → “exactly 137 levels per tick” unjustified leap); Theorem DCCP-1.7 uniform phase distribution assertion. Mode 1 pattern: numerical target known → intermediate structure asserted that produces target → forward arithmetic. Third Mode 1 recurrence today (Cal #108 + #119 + #121). - Calibration #27 STANDING (promoted Sunday 2026-05-24 PM per Casey directive after 4th instance): “BST-Primary-Target Forward-Derivation Discipline” — when a numerical target is known ($1/N_{\max}$, $\pi/N_{\max}$, $6\pi^5$, etc.), intermediate derivation steps must be FORWARD from substrate structure, not BACKWARD-engineered from the target. Verification artifacts (toys) must verify a DERIVATION, not verify a predicted target value. Four Sunday instances triggered promotion: (1) Cal #108 (Cal’s own $(1/3)^{19}$ inheritance from Lyra T2467+T2468 v0.2); (2) Cal #119 (Vol 1 Ch 5 §5.2); (3) Cal #121 (Lyra #320 v0.3 DCCP-1.5/1.7 asserted-not-derived); (4) Elie self-critique on Toy 3516 (“BUILT to verify the DCCP prediction; 6/6 PASS shows internal consistency, not independent substrate physics”). - Calibration #25 STANDING (also promoted Sunday 2026-05-24 PM per Cal #124 cross-promotion + Keeper acceptance via audit-chain delegation): “Falsifier-Threshold Discipline” — empirical tests must specify $3\sigma/5\sigma$ statistical thresholds. Operational anchor: Elie Toy 3520 photon-pair thresholds (75K probable 2σ / 169K publishable 3σ / 469K discovery 5σ ; 12-18 month SP-30-1 program scope). - Cal #123 (Toy 3516+3520 empirical-leg cold-read): both toys are MODEL SELF-CONSISTENCY CHECKS, NOT empirical-leg verification. Three-route convergence claim structurally weaker than morning’s framing. - Cal #124 (Lyra #322 v0.2 cold-read): PASS at FRAMEWORK tier; #27 flag for v0.3+ (DCCP-A_sub inherits #320 Lemma DCCP-1.5). - Elie Toy 3521 DERIVATION TEST HARNESS (Cal #27 compliant): framework accepts ANY p, tests scaling not

target; 4 mechanisms side-by-side; awaits Lyra v0.4 `lyra_v04_interface_stub()`.
7/7 PASS. - Cal #125 (Lyra #320 v0.4 + #321 v0.2 + #322 v0.3 + Toy 3521 cold-read): PASS at FRAMEWORK tier across all 4 artifacts. Lyra exemplary Calibration #27 absorption: v0.4 mechanism shift real (T2447 RIGOROUSLY CLOSED + T2476 RATIFIED upstream; Mode 1 vulnerability localized to single assumption); §3 explicit substrate-derived vs INHERITED distinction; #322 §3.1 Gap 3.1 explicit Cal #121 retraction absorption; #321 §3.3 honest sketch-grade tagging. Minor flag: #322 §2.2 forward-references “T2467+T2468 v0.3 Section 13” not yet filed (address at #322 v0.4). - Cal #126 (Lyra #320 v0.5 cold-read with Grace INV-5123 Dirac anchor closure): PASS at FRAMEWORK-PLUS (new tier between FRAMEWORK and STRUCTURALLY VERIFIED CANDIDATE). Steps 1-2 genuinely independent of BST framework (Dirac $Z\cdot\alpha=1$ textbook QED + Grace three-way identity RATIFIED). Mode 1 vulnerability confirmed v0.5 LOW-MODERATE. META-question remaining: substrate-hypothesis-selection at Step 3 (would someone independently choose Dirac scale vs §6 Candidates B/C/D without target-knowledge?). Path to STRUCTURALLY VERIFIED CANDIDATE multi-week: implement one §6 Candidate UNIQUELY selecting Dirac scale + Cal #77 alt-HSD test. New tier FRAMEWORK-PLUS defined Sunday 2026-05-24 PM by Cal #126 — Cal #66 STRUCTURALLY VERIFIED requires framework + chain + BST-internal verification complete + alt-HSD pending; v0.5 has framework + chain [?](#) but BST-internal verification of Step 3 substrate-mechanism is PARTIAL → FRAMEWORK-PLUS captures this intermediate state. - Cal Sunday total: 8 referee logs Cal #119-#126. Cal verdict: “team is exemplary today. v0.5 is the kind of work Calibration #27 STANDING discipline encourages — honest forward-derivation chain from independent upstream physics.” - Calibration #27 STANDING corollary (Grace INV-5139, Sunday 2026-05-24 PM): “When you need an independent anchor, ask the catalog lane first.” Grace literature scan IS substrate cartography Lyra can build forward from without circularity. Grace lane role formally expanded: pure-reactive catalog backbone → ALSO pure-proactive substrate cartography for forward-derivation chains. Demonstrated Sunday via Grace INV-5123 morning literature scan (unblocked queue, not Casey-asked) → Keeper Vol 12 Ch 1 absorption → Lyra #320 v0.5 forward-derivation built on same anchor. The pattern generalizes: Grace catalog INV-anchors enable Lyra forward-derivations from independent upstream sources without circularity. - Lyra #320 v0.6 + Elie Toy 3522 Sunday evening: META-vulnerability per Cal #126 quantitatively narrowed. Toy 3522 side-by-side of §6 Candidates A/B/C/D produces UNIQUE α -quantum match for Candidate A (Dirac $Z\cdot\alpha=1$); B (Casimir $\approx 16.7\%$) + C (Bergman $\approx 0.609\times\alpha$) numerically excluded; D (GF(128) $M_g \approx 1.079\times\alpha$) within 8% but experimentally discriminable from A at 5σ via Toy 3520 SPDC design. Empirical discriminability built into candidate-space → NOT pure target-fitting. Lyra v0.7+ multi-week: Candidate A Bergman kernel + K-type spectrum truncation → STRUCTURALLY VERIFIED CANDIDATE path. - Cal #127 (SP-30-1 outreach + Toy 3522 + Lyra v0.6 cold-read): SP-30-1 outreach PASS at external-survivability tier with 3 OPTIONAL polish items (Cal #50 external register softening + Dirac $Z\cdot\alpha=1$ citation + Calibration #25 link). Casey-call to send as-is or apply polish. Housekeeping flag for Lyra v0.7: §5

Lyra v0.6 “A vs C vs D all give α -scale” needs reconciliation — Toy 3522 shows only A and D give α -scale ($C = 0.609 \times \alpha$). Cal Sunday total: 9 referee logs Cal #119-#127 + Methodology Index v0.6 (Calibration #25 + #27 STANDING) + FRAMEWORK-PLUS tier coined ad-hoc. - Cal #125 methodology-functioning verdict: 4 Mode 1 catches Sunday morning → Calibration #27 STANDING → Lyra/Elie immediate absorption + exemplary v0.4/v0.2/v0.3/Toy3521 deliverables in ~1 hour. Audit-chain governance operational in same Sunday session. - Count reconciliation note (pending Cal MIndex v0.7): Cal #125 cites “24 STANDING + 1 CANDIDATE + 1 META = 26” from Methodology Index v0.6 (before Keeper acceptance of #25). After Sunday PM #25 acceptance via audit-chain delegation, current state is 25 STANDING + 1 CANDIDATE (#26) + 1 META (#99) = 27 numbered methodology layers. Awaiting Cal MIndex v0.7 update to align counts.

Methodology stack now 25 STANDING + 1 CANDIDATE (#26 Readiness-Label) + 1 META (#99) per Cal Methodology Index v0.6 (was 21 + 2 before Sunday afternoon; #25 + #27 both promoted; #26 remains CANDIDATE pending further operational instance).

10.4 Calibration as discipline accretion

Each calibration: ~5 min cost to identify + file. Each calibration: prevents ~hours-to-weeks of downstream drift. Net leverage: ~100-1000x.

This is why the team invests in calibration discipline: pays for itself many times over.

10.5 Growth rate

~1 calibration / week during active research.

Casey supports this: “calibrations are strength.”

10.6 K-audit anchors

- 22 Cal calibrations (notes/cal_referee_logs/)
- 18-layer methodology stack
- Quaker discipline (Vol 15 Ch 7) — corrections are strength

Level 3 — 5th-grader accessibility

Calibration stack: 22 Cal standing rules + 18 methodology layers accumulated from past failures. Each calibration: ~5 min cost, prevents hours of drift, 100-1000x leverage. Growth: ~1/week. Casey says: “calibrations are strength.”

What comes next

Chapter 11 develops how to continue the work.

Where to look this up

- BST: [notes/cal_referee_logs/](#)
- Vol 15 Ch 7 (Quaker discipline)